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Training Embodied Agents with Large Foundation Models

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ABSTRACT

This report introduces a framework for training embodied agents to generate realistic human motion using large foundation models. To address limitations of traditional models reliant on predefined inputs, the proposed pipeline incorporates first-person video, text descriptions, and initial poses to dynamically generate motion. Large Language Models (LLMs) are used to create structured inputs for the Controllable Human-Object Interaction Synthesis (CHOIS) model, which synthesizes realistic human motion datasets.

The methodology involves data acquisition for training a generative model and leveraging the CHOIS model to prepare the training set for the generative model. Initial experiments highlight challenges such as 3D reconstruction quality and environmental complexity during data collection, as well as the results of the output of the CHOIS model. Future work will focus on large-scale data acquisition, data processing, model training and evaluation.

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LIST OF ABBREVIATION

Abbreviation	Definition
AMASS	Archive of Motion Capture as Surface Shapes
CHOIS	Controllable Human-Object Interaction Synthesis
CCS	Camera Coordinate System
DPM	Diffusion Probabilistic Model
LLM	Large Language Model
MoCap	Motion Capture
RGBD	Red Green Blue-Depth
RNN	Recurrent Neural Network
SMPL	Skinned Multi-Person Linear
WCS	World Coordinate System

1. Introduction

1.1 Background

In recent years, advancements in deep learning, computer vision, and computer graphics have significantly accelerated the development of humanoid robots and their applications. Current human motion generation models rely heavily on human-provided inputs, such as initial human and object states, text descriptions, and waypoints [1]. While these inputs facilitate the generation of realistic motion, key limitations arise in their practicality and applicability.

Specifically, initial human and object states, as well as waypoints, are typically defined using 3D construction software. This reliance on predefined inputs restricts the usability of such models in real-world settings, where embodied agents must operate autonomously without access to predefined waypoints or meticulously constructed environments. To address this challenge, there is a growing need for systems capable of generating human motion dynamically, using contextual information derived directly from the agent's environment, without requiring manual intervention or pre-specified inputs.[1]

1.2 Objectives

To address this gap, the goal is to train a human motion generation model that can leverage first-person video, text descriptions, and an initial pose to predict realistic human motion. However, due to unexpected delays in the data collection process, an alternative approach has been adopted to create a synthetic training dataset for the motion generation model. Large Language Models(LLMs) in this case can be applied in generate a specific-task input of other models [2].

The objective of this project is to use an LLM to generate structured inputs—such as text descriptions, initial poses, and waypoints—for the CHOIS model. These inputs will enable CHOIS to generate synthetic human motion data, which will serve as a training dataset for our human motion generation model. This pipeline leverages the generative capabilities of LLMs to streamline the input creation process and ensure scalability, while CHOIS provides the corresponding motion outputs. Ultimately, this approach aims to overcome data collection challenges and advance the training of a motion generation model capable of producing realistic human motion from first-person video, text descriptions, and an initial pose.

1.3 Report Outline

The report is structured as follows: the literature review discusses existing methods and the role of LLMs in synthetic data generation; the methodology outlines the data collection process and the integration of CHOIS and LLMs; the experiments section presents initial results and challenges; the plan for remaining work focuses on data acquisition and evaluation; and the conclusion summarizes the contributions and future directions.

2. Literature Review

2.1 Diffusion Models

Diffusion models have emerged as a powerful class of generative models, characterized by their ability to iteratively transform noise into structured data. These models have shown their performance in various tasks, including image generation, speech synthesis, and molecular design. Diffusion models were introduced by Sohl-Dickstein et al. as generative frameworks based on stochastic processes, where data is gradually corrupted by noise and then reconstructed [3]. Early applications focused on image generation, but their potential for sequence modeling, including pose prediction, has only recently been explored. Traditional methods for pose prediction relied on recurrent neural networks (RNNs) or attention-based transformers, but diffusion models offer a probabilistic approach that captures the inherent uncertainty in predicting future poses.

Recent work has demonstrated the efficacy of diffusion models in human pose prediction. For instance, HumanMotionDiffusion leverages conditional diffusion processes to generate realistic and diverse motion sequences from incomplete or noisy inputs [4]. Diffusion models address the challenge of multimodal motion prediction by learning a distribution over possible future poses, rather than deterministic outputs. Additionally, MotionDiffuse introduced a framework for pose prediction conditioned on contextual signals such as past motion, text descriptions, or action labels [5]. By combining diffusion with temporal and spatial attention mechanisms, these models achieve state-of-the-art accuracy in predicting complex, multi-frame motion sequences. Advances in optimization, such as improved noise schedules and efficient sampling techniques, have further enhanced the applicability of diffusion models to real-time pose prediction scenarios.

2.2 Human motion Generation

Human motion generation refers to the process of synthesizing realistic movements of human bodies in a variety of contexts. Early approaches relied on blended to generate new animations. While MoCap systems produced highly realistic results, they were limited by the need for expensive equipment and predefined scenarios, making them unsuitable for dynamic or interactive environments.

Recent advancements in machine learning, particularly deep learning, have enabled data-driven approaches to motion generation. These methods focus on learning patterns from large datasets of human motion, allowing for more flexibility and generalization beyond pre-recorded sequences. Diffusion model especially demonstrates an outstanding result in motion generation [5], such as Interdiff [6], MDM [4] and CHOIS [1]. While these models rely on human assistance like text description and waypoints. The primary limitation of current motion generation models is their inability to operate autonomously in real-world environments. Unlike virtual settings, real-world scenarios are dynamic and unpredictable, with no predefined waypoints or meticulously constructed object states. This restricts the usability of models like CHOIS, MDM, and InterDiff outside of 3D software.

To address the limitations of current approaches, our proposed model introduces a novel method for human motion generation by integrating first-person video as an additional input. Unlike traditional models that rely solely on text descriptions and initial poses [1, 4, 6], our model leverages visual context from the robot's perspective to dynamically generate motion.

2.3 Applications for LLMs in Synthetic Data Generation

In recent years, Large Language Models (LLMs) have demonstrated immense potential in generating synthetic data across various domains [7-9]. Synthetic data refers to artificially generated data that mimics real-world data in structure and behavior. It is widely regarded for its low cost, efficiency, and scalability, making it a valuable resource for piloting new experiments or training machine learning models. By using synthetic data, researchers can accelerate development cycles while ensuring ethical and practical compliance with data-related constraints [10].

The CHOIS model has been selected for synthetic data generation in this study, owing to its advantages of high contact accuracy and reduced penetration issues over baseline models [1]. By leveraging CHOIS, the generated motion sequences can provide realistic and physically plausible training data for motion generation models.

2.4 World Coordinate System (WCS) And Camera

Coordinate System (CCS)

The World Coordinate System (WCS) and the Camera Coordinate System (CCS) are fundamental concepts in computer **vision and** robotics. The WCS defines a global, fixed reference frame for objects in a 3D space, while the CCS represents the local coordinate frame of a camera used to capture the scene. Transformations between these coordinate systems are critical for tasks such as camera calibration, 3D reconstruction, and object localization. This review explores the theoretical foundations, practical implementations, and key applications of these coordinate systems.

2.4.1 WCS

The WCS is a global reference system used to define the positions and orientations of objects in a 3D environment. It is typically defined by an origin and three orthogonal axes: X, Y, and Z. The WCS is fixed and serves as the common reference frame for aligning different objects and cameras. As noted by Hartley and Zisserman in *Multiple View Geometry in Computer Vision*, the WCS is crucial for representing the absolute positions of objects in real-world applications [11]. In robotics, the WCS is used to define the environment where the robot operates, enabling tasks like mapping and navigation. Points in the WCS are often represented as homogenous coordinates, as seen in (1), to facilitate transformations and projections.

$$P_W = [X_W, Y_W, Z_W, 1]^T \quad (1)$$

2.4.2 CCS

The CCS is a local camera-centric frame of reference, where the origin is located at the camera's optical center, and the axes are defined relative to the camera's orientation. CCS is crucial for understanding how the camera perceives the scene. In the CCS, the Z-axis typically points in the direction of the camera's optical axis, the X-axis is horizontal, and the Y-axis is vertical. The CCS is used to represent points as they are seen from the camera's perspective, making it essential for image formation and 3D-to-2D projection. According to Ma et al. in *An Invitation to 3D Vision*, the CCS is pivotal in camera calibration, where intrinsic and extrinsic parameters are used to map 3D points from the WCS into the camera frame [12]. A point in the CCS is represented as (2), where $Z_c > 0$.

$$P_C = [X_c, Y_c, Z_c]^T \quad (2)$$

3. Methodology

The following part will describe our new model and how we apply LLM and CHOIS to generate the input of our model. Two primary challenges arise: first, transfer the real-world image to the camera image with the first-person view of humanoid; and second, use LLM to generate text description and waypoints as the inputs of CHOIS, which further generate a synthetic dataset to train our model.

3.1 Getting First-person Video

This section outlines how these inputs are prepared and processed, with a particular focus on obtaining and utilizing first-person videos. To provide the first-person video input for our model, we follow a pipeline that involves recording real-world scenes with a depth camera, using these videos to train a generative model, and rendering first-person video in 3D software by applying this generative model.

The decision to use data collection and motion generation models to create first-person videos, rather than relying on direct recording, is driven by several key factors. The primary objective is to produce videos that simulate a human’s field of view, offering a first-person perspective. While direct recording is straightforward, it involves significant challenges, including the need for extensive human effort to coordinate actors, set up scenes, and manually film, making it both costly and time-consuming—especially when covering diverse scenarios. Alternatively, by collecting human motion data (e.g., through motion capture), we can train a generative model capable of synthesizing realistic human movements. This model can then render first-person perspective videos programmatically within virtual 3D environments. This data-driven approach not only reduces costs and manual effort but also provides scalability, automation, and flexibility to accommodate a wide range of applications and scenarios.

3.1.1 Record Real-world Scenes

A RGBD camera, Orbbec Femto Bolt, is used to capture video frames with depth information. It is a depth-sensing camera commonly used for applications like human motion capture, pose estimation, and 3D scanning. Its selection for this project could be justified by several features and advantages that align with the requirements of human motion generation and data acquisition tasks.

1. High-Quality Depth Sensing

The Orbbec Femto Bolt provides high-resolution depth data that is crucial for accurate motion capture and pose estimation. Its depth precision is particularly effective for handling complex scenes, including small movements or fine details of human motion.

2. Compact and Lightweight Design

The compact and lightweight design of the Orbbec Femto Bolt makes it easy to integrate into the project setups. It can be connected to the PC by an USB-C cable for power and data. This portability is ideal for collecting motion data in diverse environments without requiring bulky or stationary equipment.

3. Wide Field of View (FoV)

The camera offers a wide field of view (120° FoV), enabling it to capture a larger area of the scene and multiple objects or body parts simultaneously. This feature is particularly useful for motion capture involving dynamic human-object interactions.

4. Infrared (IR) Depth Technology

The Orbbec Femto Bolt uses infrared depth sensing, which works well even in low-light or indoor environments. This capability ensures reliable data acquisition regardless of lighting conditions, making it versatile for use in various scenarios.

5. Real-Time Performance

The Orbbec Femto Bolt supports real-time depth sensing and processing, enabling immediate data capture and feedback. This is critical for creating training datasets for motion generation models that require temporally consistent data.

6. Support for 3D Reconstruction

The Orbbec Femto Bolt enables high-quality 3D reconstruction, allowing researchers to capture human poses and objects in 3D space with precision. This is critical for generating accurate initial states of humans and objects, as required by your motion generation model.

100 hours of video are required. Data collection consists mainly of two parts. The first part is to Capture human motion data through pose estimation. The second part is scanning the environment for 3D reconstruction. For the human motion capture part, an actor and a record are required. The actor should do a series of actions to interact with the surrounding objects. The actor should do different actions according to different environments.



Figure 1 The RGBD camera

3.1.2 Reconstruction Recorded Videos

To train the generative model, the collected data is needed to be transformed into the form of mesh. The SMPL-X [13] model is applied in this part. SMPL-X model is an advanced version of Skinned Multi-Person Linear [14] (SMPL) model. The SMPL (Skinned Multi-Person Linear) model is a widely used 3D human body model that represents human shape and pose in a parametric way [14]. The model uses shape parameters to control the overall body shape and size as well as pose parameters to represent the orientation of each joint in the body. Compared with SMPL, SMPL-X not only expresses the shape and pose of the human body, it also can express the hands and face (Figure 2). As interaction between human and object is required in the training video, SMPL-X is the better choice in this case.



Figure 2 Pictures generated by the SMPL model SMPL-X model

3.1.3 Transformation from WCS to CCS

To render the first-person video by applying the generative model, we need to transform the WCS in the 3D reconstructed world to CCS. This transformation is necessary because cameras capture the world from their own perspective [11], and many computer vision algorithms, such as 3D rendering, pose estimation, and scene understanding, operate in the CCS.

The transformation between the WCS and the CCS is central to many computer vision tasks. This transformation involves both extrinsic parameters (rotation and translation) and intrinsic parameters (camera-specific properties).

Extrinsic parameters define the position and orientation of the camera in the WCS. These are represented as:

$$P_C = R \cdot P_W + T \quad (3)$$

where R is the 3×3 rotation matrix and T is the translation vector.

Intrinsic parameters map points from the CCS to the image plane. These include the focal length, skew, and principal point, represented by the camera matrix K :

$$p = K \cdot P_C \quad (4)$$

where p is the projected 2D point in the image.

The combined transformation from WCS to CCS is often expressed in homogeneous coordinates as:

$$P_C = \begin{bmatrix} R & T \\ 0 & 1 \end{bmatrix} \cdot P_W \quad (5)$$

3.2 Use LLM and CHOIS to Generate Synthetic Dataset

This part outlines the method of employing LLM and CHOIS to efficiently generate a synthetic dataset as the training set for our model. By leveraging the text generation capabilities of LLMs and the motion generation capabilities of CHOIS, this method aims to create high-quality datasets with minimal manual effort while ensuring human-in-the-loop validation for accuracy.

3.2.1 CHOIS

Conditional Human-Object Interaction Synthesis (CHOIS) is a human motion generation model based on language descriptions and sparse object waypoints, featuring in the realization of human-object interaction synthesis [1]. By leveraging a conditional diffusion model, CHOIS generates synchronized human and object motions that align with the provided language inputs while ensuring physical and contextual coherence (Figure 3). Compared with other generation models like Interdiff [6], MDM [4], Lin-OMOMO [15] and Pred-OMOMO [15], CHOIS outperforms these baseline methods in condition matching and human motion quality [1]. Also, CHOIS demonstrates improved contact accuracy and reduced penetration issues [1] in FullBodyManipulation dataset [15] and 3D-FUTURE dataset [16].

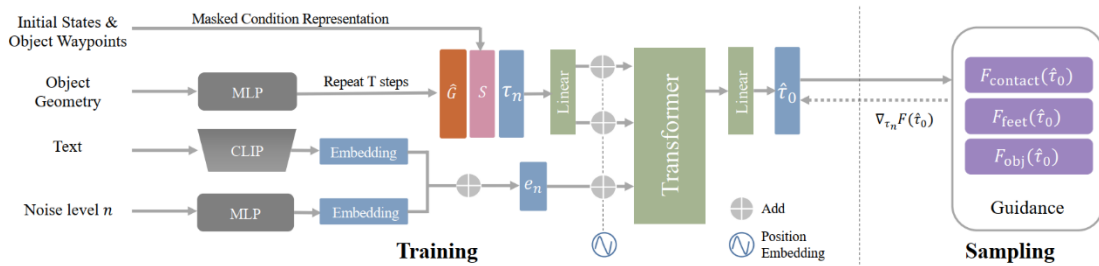


Figure 3 Workflow of CHOIS model

3.2.2 Use LLMs to Generate Inputs for CHOIS

The inputs of CHOIS consists of a language description, object geometry, initial object and human states, and sparse object waypoints [1]. To acquire components, a collection of images which a humanoid in environments with some object that can interact is required. The collection of images also functions as the initial object and human states. These images are provided as the inputs for LLM. Prompts will be provided to the LLM to generate description of potential tasks for humanoids as well as the potential waypoints conditioned on the content of the input images. With all the inputs, CHOIS can be applied to generate videos of the training dataset. Figure 4 shows the sample action generated by GPT-4o.

```
1. Sit down at the computer;  
2. Stand up from the computer;  
3. Type on the keyboard;  
4. Use the mouse to navigate the computer;  
5. Write on the whiteboard;  
6. Erase the whiteboard;  
7. Adjust the computer screen;  
8. Plug in a device to the computer;  
9. Turn on the computer;  
10. Turn off the computer;  
11. Move the keyboard closer;  
12. Move the mouse closer;  
13. Stand up to stretch;  
14. Walk over to the whiteboard to read something;  
15. Point at something on the whiteboard;  
16. Lean over to see the computer screen better;  
17. Reach for a pen or marker to use on the whiteboard;  
18. Take a picture of the whiteboard with a phone;  
19. Organize cables or wires on or around the desk;  
20. Adjust the lighting in the room.
```

Figure 4 Sample actions of the actor in a study room.

4. Experiments

The experiments mainly consist of two parts. The first part acquisition of first-person videos and the comparison of the videos with videos generated by another dataset. The second part is training video generation by CHOIS.



Figure 5 The actor and the recorder were doing data collection.

4.1 Acquisition of First-Person Video

A sample data collection was conducted in a study room area in Innovation Wing, the University of Hong Kong (Figure 5). According to the reconstruction model of the recording video (Figure 6), several issues were identified that require immediate attention and improvement. One of the primary challenges was the lack of clarity in the reconstructed 3D images, which impacted on the accuracy of capturing interactions between humans and objects. This limitation may hinder the model's ability to simulate realistic human-object interactions effectively.

The issues with the reconstructed images can be attributed to several factors. Firstly, the environment was too complex, with an excessive number of objects in space. This high level of clutter introduced precision and accuracy loss during the process of converting recorded videos into 3D reconstructions. Simplifying the environment by reducing the number of objects may help address this issue. Secondly, the presence of thin-column shaped objects (Figure 5), such as long footstools, posed challenges for the reconstruction software. These thin, elongated objects were not accurately captured, likely due to the difficulty the software has in processing slender profiles. Avoiding such objects in future setups could lead to more accurate reconstructions. Thirdly, reflective surfaces such as computer monitors and whiteboards further reduced reconstruction accuracy by causing light reflections. These reflections interfered with the software's ability to capture clear and precise images. Covering reflective surfaces with non-reflective materials would help mitigate this problem and improve the quality of the reconstructed images.



Figure 6 A sample of 3D reconstruction image generated according to the recorded videos.

4.2 Modifies on CHOIS Program

The release version of CHOIS requires several modifications before the code can be executed. This section provides a detailed analysis of the issues encountered during the execution process and outlines the methods implemented to resolve these challenges.

4.2.1 Bug 1

The first issue was encountered during executing the `trainer_chois.py` (Figure 7). According to the error report, this issue was triggered due to the miss of module “`human_body_prior.body_model`”. However, this module already existed in the root path.

This issue was solved by changing the import path for the error location in the `human_body_prior` module, from

`human_body_prior.src.human_body_prior.body_model.body_model` import `BodyModel`.

```
Traceback (most recent call last):
  File "trainer_chois.py", line 30, in
    from manip.data.cano_traj_dataset import CanoObjectTrajDataset, quat_ik_torch, quat_fk_torch
  File "/data2/zty/chois_release/manip/data/cano_traj_dataset.py", line 20, in
    from human_body_prior.body_model.body_model import BodyModel
ModuleNotFoundError: No module named 'human_body_prior.body_model'
```

Figure 7 ModuleNotFoundError

4.2.2 Bug 2

The second issue is the `AttributeError`, “`bpy.ops.import_mesh.ply`” function could not be found (Figure 8). The reason was that the version of downloaded Blender was different from the version used by CHOIS. As the README file did not mention the version of Blender, I directly downloaded the newest version. This problem was fixed by changing the version from 4.3.2 to 3.1.0.

4.2.4 Modifies on cano_traj_dataset.py

The variable `seq_count` is introduced in both functions, `extract_rest_pose_object_geometry_and_rotation` and `cal_normalize_data_input`, to impose a controlled limit on the number of sequences processed during each function's execution (Figure 11). This design choice is particularly motivated by two key considerations:

1. **Efficiency in Testing and Debugging:**

During the development and debugging phases, processing the entire dataset can be computationally expensive and time-consuming. By limiting the number of sequences processed through `seq_count`, it becomes feasible to test the functionality of the code on a smaller subset of the data. This approach ensures faster iteration cycles and enables the identification and resolution of potential issues in the pipeline without the need to repeatedly process the full dataset.

2. **Resource Optimization:**

When working with large datasets, handling all sequences simultaneously may lead to excessive memory usage or computational overhead. The use of `seq_count` allows for controlled and incremental processing of the data, ensuring that the computational resources are utilized efficiently. This is particularly beneficial for tasks like extracting rest pose object geometries or normalizing motion data, which involve computationally intensive operations such as matrix transformations and geometry processing.

```
489 def extract_rest_pose_object_geometry_and_rotation(self):
490     self.rest_pose_object_dict = {}
491     seq_count = 0
492
493     for seq_idx in self.data_dict:
494         seq_name = self.data_dict[seq_idx]['seq_name']
495         object_name = seq_name.split("_")[1]
496         if object_name in ["vacuum", "mop"]:
497             continue
498
499         if object_name not in self.rest_pose_object_dict:
500             obj_trans = self.data_dict[seq_idx]['obj_trans'][:, :, 0] # T X 3
501             obj_rot = self.data_dict[seq_idx]['obj_rot'] # T X 3 X 3
502             obj_scale = self.data_dict[seq_idx]['obj_scale'] # T
503
504             rest_verts, obj_mesh_faces, rest_pose_ori_rot, rest_pose_ori_com_pos, obj_trans_to_com_pos = \
505                 self.convert_rest_pose_obj_geometry(object_name, obj_scale, obj_trans, obj_rot)
506
507             self.rest_pose_object_dict[object_name] = {}
508             self.rest_pose_object_dict[object_name]['ori_rotation'] = rest_pose_ori_rot # 3 X 3
509             self.rest_pose_object_dict[object_name]['ori_trans'] = rest_pose_ori_com_pos # 1 X 3
510             self.rest_pose_object_dict[object_name]['obj_trans_to_com_pos'] = obj_trans_to_com_pos # 1 X 3
511
512         seq_count += 1
513         if seq_count >= 10:
514             break
515
```

```

516     def cal_normalize_data_input(self):
517         self.window_data_dict = {}
518         s_idx = 0
519         seq_count = 0
520
521         for index in self.data_dict:
522             seq_name = self.data_dict[index]['seq_name']
523
524             object name = seq_name.split(" ")[1]

```

```

516     def cal_normalize_data_input(self):
642
643         self.window_data_dict[s_idx]['rest_human_offsets'] = rest_human_offsets
644
645         s_idx += 1
646
647         seq_count += 1
648         if seq_count >= 10:
649             break
650

```

Figure 11 Modified functions

After fixing the above issues, the CHOIS can be run successfully.

4.3 Running of CHOIS model

This part will elaborate the logistics of the main code of CHOIS model and show the result of the model.

By executing the command in Figure 12, the CHOIS model can be run.

```
sh scripts/test_chois_long_seq_in_scene.sh
```

Figure 12 Command to run the CHOIS model

According to the paper of CHOIS, the inputs needed by the model consist of the text description, the initial states of the human and the object, the object geometry and sparse waypoints. In code realization, the inputs are packed into a pickle file (Figure 13). The components of the pickle file are shown in the Figure 14. The 3D scenes used are from the Replica dataset [17]. The waypoints are generated by applying Habitat-sim [18].

```

≡ cano_min_max_mean_std_data_window_120_joints24.p
≡ cano_test_diffusion_manip_window_120_joints24.p
≡ cano_train_diffusion_manip_window_120_joints24.p
≡ test_diffusion_manip_seq_joints24.p
≡ train_diffusion_manip_seq_joints24.p
≡ whole_seq_test_diffusion_manip_window_joints24_all.p

```

Figure 13 The packed inputs

```
数据类型: <class 'dict'>
数据条目数: 602
seq_name: <class 'str'>, shape: N/A
betas: <class 'numpy.ndarray'>, shape: (1, 16)
gender: <class 'numpy.ndarray'>, shape: ()
trans2joint: <class 'numpy.ndarray'>, shape: (3,)
rest_offsets: <class 'numpy.ndarray'>, shape: (24, 3)
trans: <class 'numpy.ndarray'>, shape: (143, 3)
root_orient: <class 'numpy.ndarray'>, shape: (143, 3)
pose_body: <class 'numpy.ndarray'>, shape: (143, 63)
obj_scale: <class 'numpy.ndarray'>, shape: (143,)
obj_trans: <class 'numpy.ndarray'>, shape: (143, 3, 1)
obj_rot: <class 'numpy.ndarray'>, shape: (143, 3, 3)
obj_com_pos: <class 'numpy.ndarray'>, shape: (143, 3)
```

Figure 14 test_diffusion_manip_seq_joints24.p

The CHOIS model leverages a conditional diffusion model (ObjectCondGaussianDiffusion) to predict human motion dynamics conditioned on object trajectories, language prompts, and other contextual features. The training process is managed by the Trainer class, which integrates techniques like Exponential Moving Average (EMA) for model stabilization and mixed precision training for efficiency. The model predicts object-relative transformations, global human joint positions, and orientations using a transformer-based architecture. The framework supports generating single-window or continuous long sequences by employing planned paths as waypoints, ensuring smooth transitions between frames. Additionally, it includes guidance mechanisms during the denoising process, such as feet-floor contact, hand-object interaction, and scene penetration constraints, to improve physical plausibility. For evaluation, the program computes various metrics (e.g., joint position error, contact accuracy, and penetration scores) and provides extensive visualization options, including object and human meshes rendered in Blender. Its modular design allows for integration with diverse datasets, object splits, and language-based conditions, enabling both qualitative and quantitative analysis of generated results.

Examples of visualization of generated frames could be seen in Figure 15 and Figure 16.

These two frames are from the 101st frame of the sequence of `fri_apartment_4_sub16_largebox_005_largebox_pidx_0_sample_cnt_0`. Figure 15 showed the predicted frame without a 3D scene and floor and Figure 16 showed the predicted frame top view with the sample 3D scene in Replica dataset. It can be observed that the relative positioning between the box and the human is generally satisfactory, and the contact between the foot and the floor is well-handled. However, there is noticeable penetration between the hands and the box.

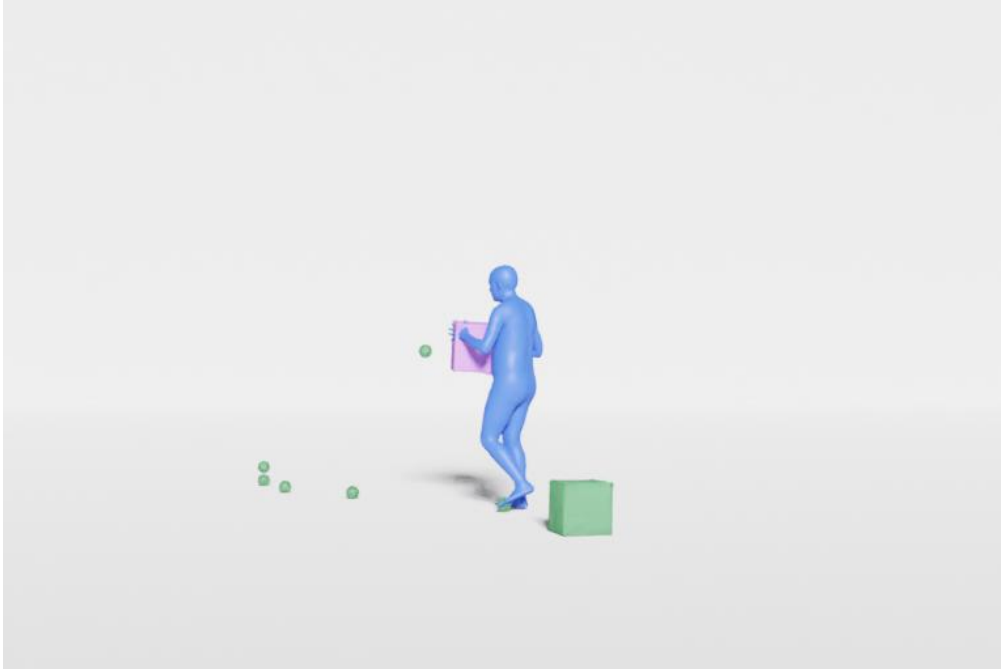


Figure 15 Predicted frame with no scene



Figure 16 The top view of predicted frame

5. Evaluation

By executing the command “sh scripts/test_chois_long_seq_in_scene.sh --for_quant_eval”, the evaluation results for each frame can be obtained. Figure 17 is a sample output of the evaluation results of the sequence

f1r_apartment_4_sub16_clothesstand_004_clothesstand_pidx_8_sample_cnt_0.

```

Current Sequence name:frl_apartment_4_sub16_clothesstand_004_clothesstand_pidx_7_sample_cnt_0
The number of sequences: 1
*****Human Motion Evaluation*****
Foot sliding jnts: 2.8545763829480046
Floor Height: 0.04681846499443054
*****Interaction Evaluation*****
Hand-Object Penetration Score: 8.822102972771972e-05
Human-Object Penetration Score: 1.3827331713400781e-05
Contact percentage: 0.9652173913043478
Scene-Human Penetration Score: 0.00013269348710309714
Scene-Object Penetration Score: 0.0007801500032655895
*****Condition Following Evaluation*****
Start obj_com_pos err: 21.10356092453003, End obj_com_pos err: 92.91858226060867
waypoints xy err: 46.01746425032616

```

Figure 17 Evaluation output

To evaluate the actual outcomes of the CHOIS model, four aspects were selected to compare with the data in the paper. It includes foot sliding (FS), contact percentage (C%), hand-object penetration score (P_{hand}) and waypoint errors (T_{xy}).

Table 1 Evaluation output comparison

	FS	C%	P_{hand}	T_{xy}
Paper data	0.35	0.54	0.59	2.87
Actual data	5.46	0.73	0.028	46.02

5.1 Foot Sliding

The foot sliding (FS) metric shows a significant increase from the paper's value of 0.35 to your actual result of 5.46. This indicates a major issue with maintaining stable or natural motion in your system compared to the values reported in the paper. Such a large increase in FS suggests instability, likely caused by deficiencies in motion planning, execution, or environmental factors. Possible reasons for this discrepancy include issues with the motion planning algorithm, poor friction modeling, or inconsistencies in surface properties. Environmental conditions, such as irregularities in the ground or improper contact simulation, could also contribute to this result. To address this, it is important to evaluate the motion planning system, refine ground contact and friction handling, and ensure that the environmental conditions align with those described in the paper.

5.2 Contact Percentage

The contact percentage (C%) metric has increased moderately from 0.54 in the paper to 0.73 in your actual results. This indicates improved consistency in maintaining effective contact between the object and the intended surface. The 35% improvement in C% is a positive

outcome, as it suggests better control over the interaction between the system and its environment. This improvement could be the result of enhanced grasping or manipulation strategies, better contact modeling, or improvements in hardware. While the increase in C% is promising, it is essential to verify that this improvement is consistent across various scenarios and does not come at the expense of other metrics, such as foot sliding (FS) or waypoint errors (T_xy).

5.3 Hand-Object Penetration Score

The hand-object penetration score (P_hand) has decreased drastically from 0.59 in the paper to 0.028 in your actual results, representing a significant improvement. This reduction indicates that your system is far more effective at avoiding collisions or interpenetration between the hand and the object, leading to more accurate and realistic physical interactions. This improvement could be attributed to better collision detection algorithms, enhanced control strategies for hand-object interaction, or improved modeling of object boundaries. This is a very positive result, as it demonstrates that your system is achieving higher levels of precision in handling objects. However, you should confirm whether this improvement holds across varied conditions and ensure that the reduction in P_hand does not negatively impact other metrics, such as trajectory accuracy or contact consistency.

5.4 Waypoint Errors

The waypoint errors (T_xy) metric shows a dramatic increase from 2.87 in the paper to 46.02 in your actual results. This represents a significant problem, as waypoint errors reflect the system's ability to accurately follow planned trajectories. A 16-fold increase suggests that your system is struggling with trajectory planning or execution. Potential causes include inadequate path-planning algorithms, poor control mechanisms, environmental disturbances, or hardware limitations, such as low sensor or actuator precision. This large discrepancy implies severe inefficiencies or bottlenecks in the trajectory planning pipeline, which needs immediate attention. To resolve this, you should analyze the trajectory planning and execution processes, identify specific sources of error, and optimize the system to reduce inaccuracies. Additionally, testing the system in controlled environments may help isolate the issues and provide clarity on whether the problems stem from environmental conditions or system design.

5.5 Summary and Recommendations

Overall, the actual results show mixed outcomes when compared to the paper. On the positive side, contact percentage (C%) has improved significantly, indicating better consistency in interactions, and the hand-object penetration score (P_hand) has been drastically reduced, showing enhanced precision in avoiding penetration errors. However, there are notable challenges, particularly with foot sliding (FS) and waypoint errors (T_xy). The sharp increase in FS suggests instability and unnatural motion, while the dramatic rise in T_xy highlights significant inefficiencies in trajectory planning and execution. These discrepancies between the paper's reported values and the actual results suggest that further optimization and refinement of the system are necessary.

To address the issues highlighted by the analysis, several steps can be taken. First, the problem of foot sliding should be prioritized by improving ground contact modeling, refining motion planning algorithms, and ensuring accurate friction handling. Second, the waypoint error metric must be reduced by identifying bottlenecks in the trajectory planning pipeline, refining path-planning algorithms, and improving hardware precision where necessary. Third, the improvements in contact percentage and hand-object penetration should be leveraged to enhance the system further. Testing the system under varied conditions will help confirm whether these improvements are consistent. Finally, it is important to compare the experimental conditions, methodologies, and setups in your implementation with those described in the paper to identify any critical differences that may be contributing to the discrepancies. By addressing these issues systematically, the overall performance of the system can be optimized to align more closely with the paper's reported results.

6. Remaining Work

The large-scale data acquisition is currently underway. Once the dataset is obtained, a systematic process will be implemented to transform the collected data into first-person video data. This step will ensure that the data is properly formatted and aligned with the requirements for subsequent model training and evaluation.

In terms of the training set generation, inputs will be carefully prepared and provided to the CHOIS model to facilitate the generation of videos. These generated videos, along with first-person video data, will then be fed into the diffusion model for further processing. Finally, a comprehensive evaluation will be conducted to assess the performance and effectiveness of the model, using metrics and benchmarks relevant to the task. This structured pipeline will ensure

a seamless flow of data from acquisition to evaluation, supporting the development of robust and high-quality first-person video generation.

The current work is behind schedule.

7. Conclusion

This project explored the development of a robust framework for generating first-person videos using the CHOIS model and a diffusion-based approach. The process began with systematic data acquisition, where raw data was collected and trained a generative model to render first-person video in 3D software. A pipeline was established to prepare inputs for the CHOIS model, enabling the generation of realistic and semantically meaningful video outputs. These generated videos, along with the collected first-person data, were integrated into the diffusion model to ensure high-quality video generation aligned with human-object interaction scenarios. Future work will focus on expanding the dataset, processing the collected dataset, refining the models, and improving the scalability and generalization of the framework to handle more diverse and complex scenarios.

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